414. $\left(\frac{5}{9}+\frac{1}{6}\right)\div\left(\frac{2}{3}-\frac{1}{2}\right)$

415. $\left(\frac{3}{4}+\frac{1}{6}\right)\div\left(\frac{5}{8}-\frac{1}{3}\right)$

In the following exercises, evaluate the given expression. Express your answers in simplified form, using improper fractions if necessary.

416. $x+\frac{1}{2}$ when
ⓐ $x=-\frac{1}{8}$ ⓑ $x=-\frac{1}{2}$

417. $x+\frac{2}{3}$ when
ⓐ $x=-\frac{1}{6}$ ⓑ $x=-\frac{5}{3}$

418. $x+\left(-\frac{5}{6}\right)$ when
ⓐ $x=\frac{1}{3}$ ⓑ $x=-\frac{1}{6}$

419. $x+\left(-\frac{11}{12}\right)$ when
ⓐ $x=\frac{11}{12}$ ⓑ $x=\frac{3}{4}$

420. $x-\frac{2}{5}$ when
ⓐ $x=\frac{3}{5}$ ⓑ $x=-\frac{3}{5}$

421. $x-\frac{1}{3}$ when
ⓐ $x=\frac{2}{3}$ ⓑ $x=-\frac{2}{3}$

422. $\frac{7}{10}-w$ when
ⓐ $w=\frac{1}{2}$ ⓑ $w=-\frac{1}{2}$

423. $\frac{5}{12}-w$ when
ⓐ $w=\frac{1}{4}$ ⓑ $w=-\frac{1}{4}$

424. $4p^2q$ when $p=-\frac{1}{2}$ and $q=\frac{5}{9}$

425. $5m^2n$ when $m=-\frac{2}{5}$ and $n=\frac{1}{3}$

426. $2x^2y^3$ when $x=-\frac{2}{3}$ and $y=-\frac{1}{2}$

427. $8u^2v^3$ when $u=-\frac{3}{4}$ and $v=-\frac{1}{2}$

428. $\frac{u+v}{w}$ when $u=-4, v=-8, w=2$

429. $\frac{m+n}{p}$ when $m=-6, n=-2, p=4$

430. $\frac{a+b}{a-b}$ when $a=-3, b=8$

431. $\frac{r-s}{r+s}$ when $r=10, s=-5$

Everyday Math

432. **Decorating** Laronda is making covers for the throw pillows on her sofa. For each pillow cover, she needs $\frac{3}{16}$ yard of print fabric and $\frac{3}{8}$ yard of solid fabric. What is the total amount of fabric Laronda needs for each pillow cover?

433. **Baking** Vanessa is baking chocolate chip cookies and oatmeal cookies. She needs $1\frac{1}{4}$ cups of sugar for the chocolate chip cookies, and $1\frac{1}{8}$ cups for the oatmeal cookies How much sugar does she need altogether?

Writing Exercises

434. Explain why it is necessary to have a common denominator to add or subtract fractions.

435. Explain how to find the LCD of two fractions.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
add and subtract fractions with different denominators.			
identify and use fraction operations.			
use the order of operations to simplify complex fractions.			
evaluate variable expressions with fractions.			

ⓑ *After looking at the checklist, do you think you are well prepared for the next section? Why or why not?*

4.6 Add and Subtract Mixed Numbers

Learning Objectives

By the end of this section, you will be able to:

- Model addition of mixed numbers with a common denominator
- Add mixed numbers with a common denominator
- Model subtraction of mixed numbers
- Subtract mixed numbers with a common denominator
- Add and subtract mixed numbers with different denominators

☑ **BE PREPARED** 4.14 Before you get started, take this readiness quiz.

Draw a model of the fraction $\frac{7}{3}$.

If you missed this problem, review Example 4.6.

☑ **BE PREPARED** 4.15 Change $\frac{11}{4}$ to a mixed number.

If you missed this problem, review Example 4.9.

☑ **BE PREPARED** 4.16 Change $3\frac{1}{2}$ to an improper fraction.

If you missed this problem, review Example 4.11.

Model Addition of Mixed Numbers with a Common Denominator

So far, we've added and subtracted proper and improper fractions, but not mixed numbers. Let's begin by thinking about addition of mixed numbers using money.

If Ron has 1 dollar and 1 quarter, he has $1\frac{1}{4}$ dollars.

If Don has 2 dollars and 1 quarter, he has $2\frac{1}{4}$ dollars.

What if Ron and Don put their money together? They would have 3 dollars and 2 quarters. They add the dollars and add the quarters. This makes $3\frac{2}{4}$ dollars. Because two quarters is half a dollar, they would have 3 and a half dollars, or $3\frac{1}{2}$ dollars.

$$\begin{array}{r} 1\frac{1}{4} \\ +2\frac{1}{4} \\ \hline \end{array}$$

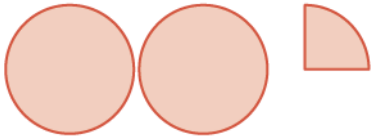
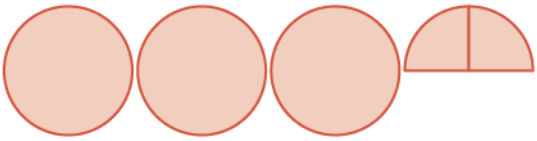
$$3\frac{2}{4} = 3\frac{1}{2}$$

When you added the dollars and then added the quarters, you were adding the whole numbers and then adding the fractions.

$$1\frac{1}{4} + 2\frac{1}{4}$$

We can use fraction circles to model this same example:

		$1\frac{1}{4} + 2\frac{1}{4}$	
Start with $1\frac{1}{4}$.	one whole and one $\frac{1}{4}$ pieces		$1\frac{1}{4}$

Add $2\frac{1}{4}$ more.	two wholes and one $\frac{1}{4}$ pieces	+ 	$+\ 2\frac{1}{4}$
The sum is:	three wholes and two $\frac{1}{4}$'s		$3\frac{2}{4} = 3\frac{1}{2}$

MANIPULATIVE MATHEMATICS

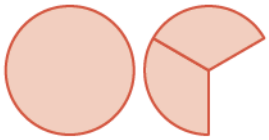
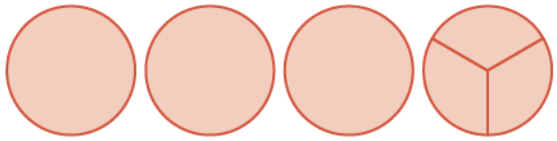
Doing the Manipulative Mathematics activity "Model Mixed Number Addition/Subtraction" will help you develop a better understanding of adding and subtracting mixed numbers.

EXAMPLE 4.81

Model $2\frac{1}{3} + 1\frac{2}{3}$ and give the sum.

Solution

We will use fraction circles, whole circles for the whole numbers and $\frac{1}{3}$ pieces for the fractions.

two wholes and one $\frac{1}{3}$		$2\frac{1}{3}$
plus one whole and two $\frac{1}{3}$s	+ 	$+1\frac{2}{3}$
sum is three wholes and three $\frac{1}{3}$s		$3\frac{3}{3} = 4$

This is the same as 4 wholes. So, $2\frac{1}{3} + 1\frac{2}{3} = 4$.

TRY IT 4.161 Use a model to add the following. Draw a picture to illustrate your model.

$1\frac{2}{5} + 3\frac{3}{5}$

TRY IT 4.162 Use a model to add the following. Draw a picture to illustrate your model.

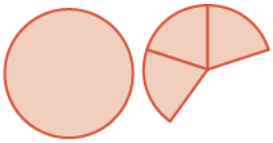
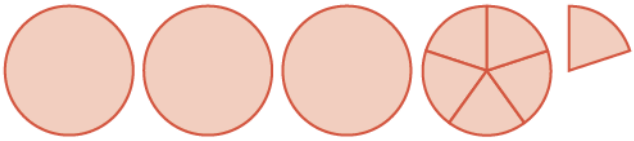
$2\frac{1}{6}+2\frac{5}{6}$

EXAMPLE 4.82

Model $1\frac{3}{5}+2\frac{3}{5}$ and give the sum as a mixed number.

Solution

We will use fraction circles, whole circles for the whole numbers and $\frac{1}{5}$ pieces for the fractions.

one whole and three $\frac{1}{5}$s		$1\frac{3}{5}$
plus two wholes and three $\frac{1}{5}$s.		$+2\frac{3}{5}$
sum is three wholes and six $\frac{1}{5}$s		$3\frac{6}{5}=4\frac{1}{5}$

Adding the whole circles and fifth pieces, we got a sum of $3\frac{6}{5}$. We can see that $\frac{6}{5}$ is equivalent to $1\frac{1}{5}$, so we add that to the 3 to get $4\frac{1}{5}$.

TRY IT 4.163 Model, and give the sum as a mixed number. Draw a picture to illustrate your model.

$2\frac{5}{6}+1\frac{5}{6}$

TRY IT 4.164 Model, and give the sum as a mixed number. Draw a picture to illustrate your model.

$1\frac{5}{8}+1\frac{7}{8}$

Add Mixed Numbers

Modeling with fraction circles helps illustrate the process for adding mixed numbers: We add the whole numbers and add the fractions, and then we simplify the result, if possible.

HOW TO

Add mixed numbers with a common denominator.

Step 1. Add the whole numbers.

Step 2. Add the fractions.

Step 3. Simplify, if possible.

EXAMPLE 4.83

Add: $3\frac{4}{9} + 2\frac{2}{9}$.

Solution

	$3\frac{4}{9} + 2\frac{2}{9}$
Add the whole numbers.	$\begin{array}{r} 3\frac{4}{9} \\ +\,2\frac{2}{9} \\ \hline 5 \end{array}$
Add the fractions.	$\begin{array}{r} 3\frac{4}{9} \\ +\,2\frac{2}{9} \\ \hline 5\frac{6}{9} \end{array}$
Simplify the fraction.	$\begin{array}{r} 3\frac{4}{9} \\ +\,2\frac{2}{9} \\ \hline 5\frac{6}{9} = 5\frac{2}{3} \end{array}$

TRY IT 4.165 Find the sum: $4\frac{4}{7} + 1\frac{2}{7}$.

TRY IT 4.166 Find the sum: $2\frac{3}{11} + 5\frac{6}{11}$.

In Example 4.83, the sum of the fractions was a proper fraction. Now we will work through an example where the sum is an improper fraction.

EXAMPLE 4.84

Find the sum: $9\frac{5}{9} + 5\frac{7}{9}$.

Solution

	$9\frac{5}{9} + 5\frac{7}{9}$
Add the whole numbers and then add the fractions.	$\begin{array}{r} 9\frac{5}{9} \\ +\,5\frac{7}{9} \\ \hline 14\frac{12}{9} \end{array}$
Rewrite $\frac{12}{9}$ as a mixed number.	$14 + 1\frac{3}{9}$

Add.	$15\frac{3}{9}$
Simplify.	$15\frac{1}{3}$

TRY IT 4.167 Find the sum: $8\frac{7}{8}+7\frac{5}{8}$.

TRY IT 4.168 Find the sum: $6\frac{7}{9}+8\frac{5}{9}$.

An alternate method for adding mixed numbers is to convert the mixed numbers to improper fractions and then add the improper fractions. This method is usually written horizontally.

EXAMPLE 4.85

Add by converting the mixed numbers to improper fractions: $3\frac{7}{8}+4\frac{3}{8}$.

Solution

	$3\frac{7}{8}+4\frac{3}{8}$
Convert to improper fractions.	$\frac{31}{8}+\frac{35}{8}$
Add the fractions.	$\frac{31+35}{8}$
Simplify the numerator.	$\frac{66}{8}$
Rewrite as a mixed number.	$8\frac{2}{8}$
Simplify the fraction.	$8\frac{1}{4}$

Since the problem was given in mixed number form, we will write the sum as a mixed number.

TRY IT 4.169 Find the sum by converting the mixed numbers to improper fractions: $5\frac{5}{9}+3\frac{7}{9}$.

TRY IT 4.170 Find the sum by converting the mixed numbers to improper fractions: $3\frac{7}{10}+2\frac{9}{10}$.

Table 4.2 compares the two methods of addition, using the expression $3\frac{2}{5}+6\frac{4}{5}$ as an example. Which way do you

prefer?

Mixed Numbers	Improper Fractions
$3\frac{2}{5}$ $+6\frac{4}{5}$ $9\frac{6}{5}$ $9+\frac{6}{5}$ $9+1\frac{1}{5}$ $10\frac{1}{5}$	$3\frac{2}{5}+6\frac{4}{5}$ $\frac{17}{5}+\frac{34}{5}$ $\frac{51}{5}$ $10\frac{1}{5}$

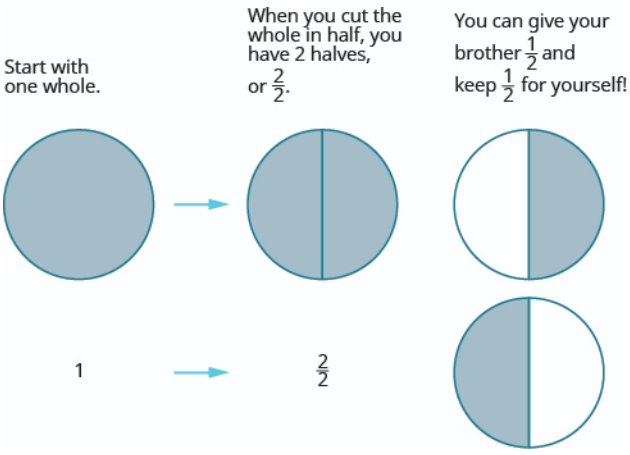
Table 4.2

Model Subtraction of Mixed Numbers

Let's think of pizzas again to model subtraction of mixed numbers with a common denominator. Suppose you just baked a whole pizza and want to give your brother half of the pizza. What do you have to do to the pizza to give him half? You have to cut it into at least two pieces. Then you can give him half.

We will use fraction circles (pizzas!) to help us visualize the process.

Start with one whole.



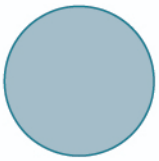
Algebraically, you would write:

$$1 \quad \frac{2}{2} \quad \frac{2}{2}$$
$$-\frac{1}{2} \rightarrow -\frac{1}{2} \rightarrow -\frac{1}{2}$$
$$\frac{1}{2}$$

EXAMPLE 4.86

Use a model to subtract: $1-\frac{1}{3}$.

Solution

	Model	Math Notation
Rewrite vertically. Start with one whole.		1 $-\frac{1}{3}$ ___
Since $\frac{1}{3}$ has denominator 3, cut the whole into 3 pieces. The 1 whole becomes $\frac{3}{3}$.		$\frac{3}{3}$ $-\frac{1}{3}$ ___
Take away $\frac{1}{3}$. There are $\frac{2}{3}$ left.		$\frac{3}{3}$ $-\frac{1}{3}$ $\frac{2}{3}$

TRY IT 4.171 Use a model to subtract: $1-\frac{1}{4}$.

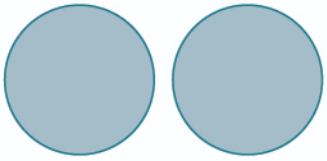
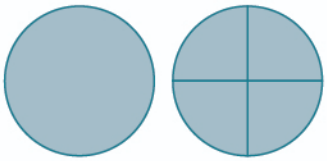
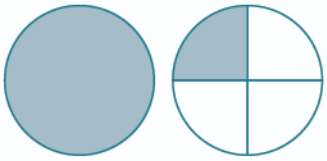
TRY IT 4.172 Use a model to subtract: $1-\frac{1}{5}$.

What if we start with more than one whole? Let's find out.

EXAMPLE 4.87

Use a model to subtract: $2-\frac{3}{4}$.

Solution

	Model	Math Notation
Rewrite vertically. Start with two wholes.		2 $-\frac{3}{4}$ ___
Since $\frac{3}{4}$ has denominator 4, cut one of the wholes into 4 pieces. You have 1 whole and $\frac{4}{4}$.		$1\frac{4}{4}$ $-\frac{3}{4}$ ___
Take away $\frac{3}{4}$. There is $1\frac{1}{4}$ left.		$1\frac{4}{4}$ $-\frac{3}{4}$ $1\frac{1}{4}$

TRY IT 4.173 Use a model to subtract: $2 - \frac{1}{5}$.

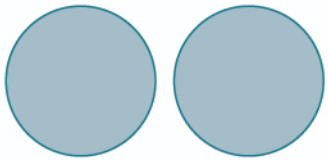
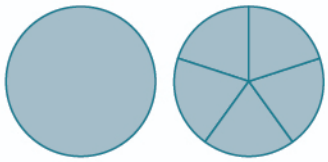
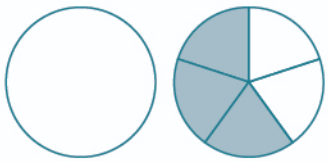
TRY IT 4.174 Use a model to subtract: $2 - \frac{1}{3}$.

In the next example, we'll subtract more than one whole.

EXAMPLE 4.88

Use a model to subtract: $2 - 1\frac{2}{5}$.

Solution

	Model	Math Notation
Rewrite vertically. Start with two wholes.		2 $-1\frac{2}{5}$
Since $\frac{2}{5}$ has denominator 5, cut one of the wholes into 5 pieces. You have 1 whole and $\frac{5}{5}$.		$1\frac{5}{5}$ $-1\frac{2}{5}$
Take away $1\frac{2}{5}$. There is $\frac{3}{5}$ left.		$1\frac{5}{5}$ $-1\frac{2}{5}$ $\frac{3}{5}$

TRY IT 4.175 Use a model to subtract: $2 - 1\frac{1}{3}$.

TRY IT 4.176 Use a model to subtract: $2 - 1\frac{1}{4}$.

What if you start with a mixed number and need to subtract a fraction? Think about this situation: You need to put three quarters in a parking meter, but you have only a $\$1$ bill and one quarter. What could you do? You could change the dollar bill into 4 quarters. The value of 4 quarters is the same as one dollar bill, but the 4 quarters are more useful for the parking meter. Now, instead of having a $\$1$ bill and one quarter, you have 5 quarters and can put 3 quarters in the meter.

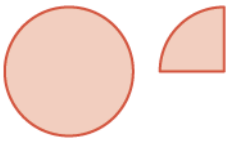
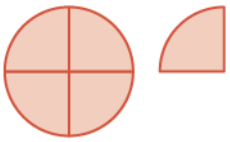
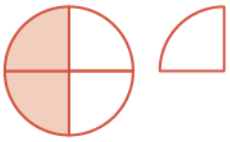
This models what happens when we subtract a fraction from a mixed number. We subtracted three quarters from one dollar and one quarter.

We can also model this using fraction circles, much like we did for addition of mixed numbers.

EXAMPLE 4.89

Use a model to subtract: $1\frac{1}{4} - \frac{3}{4}$

Solution

Rewrite vertically. Start with one whole and one fourth.		$1\frac{1}{4}$ $-\frac{3}{4}$
Since the fractions have denominator 4, cut the whole into 4 pieces. You now have $\frac{4}{4}$ and $\frac{1}{4}$ which is $\frac{5}{4}$.		$\frac{5}{4}$ $-\frac{3}{4}$
Take away $\frac{3}{4}$. There is $\frac{1}{2}$ left.		$\frac{5}{4}$ $-\frac{3}{4}$ $\frac{2}{4}=\frac{1}{2}$

TRY IT 4.177 Use a model to subtract. Draw a picture to illustrate your model.

$1\frac{1}{3}-\frac{2}{3}$

TRY IT 4.178 Use a model to subtract. Draw a picture to illustrate your model.

$1\frac{1}{5}-\frac{4}{5}$

Subtract Mixed Numbers with a Common Denominator

Now we will subtract mixed numbers without using a model. But it may help to picture the model in your mind as you read the steps.

HOW TO

Subtract mixed numbers with common denominators.

Step 1. Rewrite the problem in vertical form.

Step 2. Compare the two fractions.
- If the top fraction is larger than the bottom fraction, go to Step 3.
- If not, in the top mixed number, take one whole and add it to the fraction part, making a mixed number with an improper fraction.

Step 3. Subtract the fractions.

Step 4. Subtract the whole numbers.

Step 5. Simplify, if possible.

EXAMPLE 4.90

Find the difference: $5\frac{3}{5}-2\frac{4}{5}$.

Solution

	$5\frac{3}{5}-2\frac{4}{5}$
Rewrite the problem in vertical form.	$5\frac{3}{5}$ $-2\frac{4}{5}$
Since $\frac{3}{5}$ is less than $\frac{4}{5}$, take 1 from the 5 and add it to the $\frac{3}{5}$: $\left(\frac{5}{5}+\frac{3}{5}=\frac{8}{5}\right)$	$5\frac{3}{5} \rightarrow 4\frac{8}{5}$ $-2\frac{4}{5} \quad -2\frac{4}{5}$
Subtract the fractions.	$4\frac{8}{5}$ $-2\frac{4}{5}$ $\frac{4}{5}$
Subtract the whole parts. The result is in simplest form.	$4\frac{8}{5}$ $-2\frac{4}{5}$ $2\frac{4}{5}$

Since the problem was given with mixed numbers, we leave the result as mixed numbers.

TRY IT 4.179 Find the difference: $6\frac{4}{9}-3\frac{7}{9}$.

TRY IT 4.180 Find the difference: $4\frac{4}{7}-2\frac{6}{7}$.

Just as we did with addition, we could subtract mixed numbers by converting them first to improper fractions. We should write the answer in the form it was given, so if we are given mixed numbers to subtract we will write the answer as a mixed number.

HOW TO

Subtract mixed numbers with common denominators as improper fractions.

Step 1. Rewrite the mixed numbers as improper fractions.

Step 2. Subtract the numerators.

Step 3. Write the answer as a mixed number, simplifying the fraction part, if possible.

EXAMPLE 4.91

Find the difference by converting to improper fractions:

$9\frac{6}{11} - 7\frac{10}{11}$.

Solution

	$9\frac{6}{11} - 7\frac{10}{11}$
Rewrite as improper fractions.	$\frac{105}{11} - \frac{87}{11}$
Subtract the numerators.	$\frac{18}{11}$
Rewrite as a mixed number.	$1\frac{7}{11}$

TRY IT 4.181 Find the difference by converting the mixed numbers to improper fractions: $6\frac{4}{9} - 3\frac{7}{9}$.

TRY IT 4.182 Find the difference by converting the mixed numbers to improper fractions: $4\frac{4}{7} - 2\frac{6}{7}$.

Add and Subtract Mixed Numbers with Different Denominators

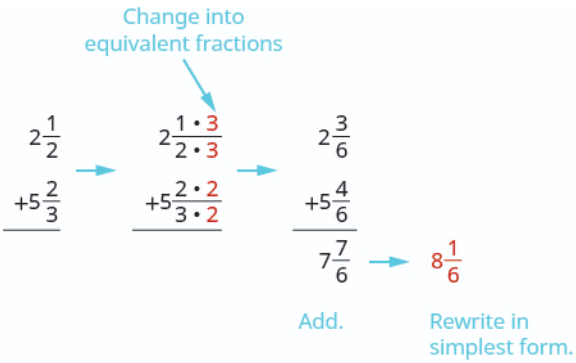
To add or subtract mixed numbers with different denominators, we first convert the fractions to equivalent fractions with the LCD. Then we can follow all the steps we used above for adding or subtracting fractions with like denominators.

EXAMPLE 4.92

Add: $2\frac{1}{2} + 5\frac{2}{3}$.

Solution

Since the denominators are different, we rewrite the fractions as equivalent fractions with the LCD, 6. Then we will add and simplify.



We write the answer as a mixed number because we were given mixed numbers in the problem.

TRY IT 4.183 Add: $1\frac{5}{6} + 4\frac{3}{4}$.

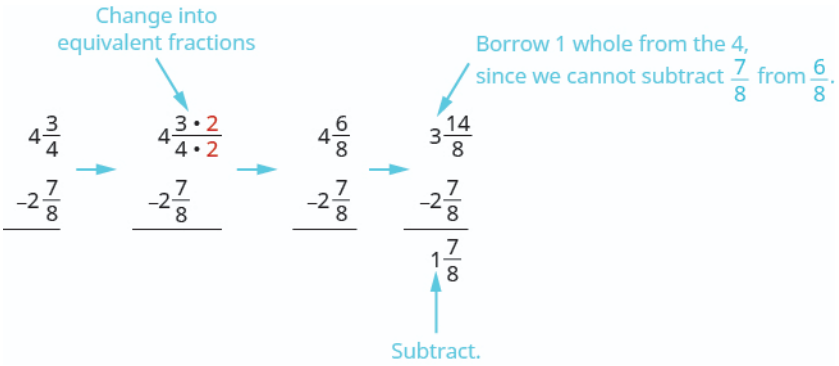
TRY IT 4.184 Add: $3\frac{4}{5} + 8\frac{1}{2}$.

EXAMPLE 4.93

Subtract: $4\frac{3}{4} - 2\frac{7}{8}$.

Solution

Since the denominators of the fractions are different, we will rewrite them as equivalent fractions with the LCD 8. Once in that form, we will subtract. But we will need to borrow 1 first.



We were given mixed numbers, so we leave the answer as a mixed number.

TRY IT 4.185 Find the difference: $8\frac{1}{2} - 3\frac{4}{5}$.

TRY IT 4.186 Find the difference: $4\frac{3}{4} - 1\frac{5}{6}$.

EXAMPLE 4.94

Subtract: $3\frac{5}{11} - 4\frac{3}{4}$.

Solution

We can see the answer will be negative since we are subtracting 4 from 3. Generally, when we know the answer will be negative it is easier to subtract with improper fractions rather than mixed numbers.

	$3\frac{5}{11} - 4\frac{3}{4}$
Change to equivalent fractions with the LCD.	$3\frac{5\cdot4}{11\cdot4} - 4\frac{3\cdot11}{4\cdot11}$ $3\frac{20}{44} - 4\frac{33}{44}$
Rewrite as improper fractions.	$\frac{152}{44} - \frac{209}{44}$
Subtract.	$-\frac{57}{44}$
Rewrite as a mixed number.	$-1\frac{13}{44}$

TRY IT 4.187 Subtract: $1\frac{3}{4} - 6\frac{7}{8}$.

TRY IT 4.188 Subtract: $10\frac{3}{7} - 22\frac{4}{9}$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Adding Mixed Numbers (https://www.openstax.org/l/24AddMixed)

Subtracting Mixed Numbers (https://www.openstax.org/l/24SubtractMixed)

SECTION 4.6 EXERCISES

Practice Makes Perfect

Model Addition of Mixed Numbers

In the following exercises, use a model to find the sum. Draw a picture to illustrate your model.

436. $1\frac{1}{5}+3\frac{1}{5}$

437. $2\frac{1}{3}+1\frac{1}{3}$

438. $1\frac{3}{8}+1\frac{7}{8}$

439. $1\frac{5}{6}+1\frac{5}{6}$

Add Mixed Numbers with a Common Denominator

In the following exercises, add.

440. $5\frac{1}{3}+6\frac{1}{3}$

441. $2\frac{4}{9}+5\frac{1}{9}$

442. $4\frac{5}{8}+9\frac{3}{8}$

443. $7\frac{9}{10}+3\frac{1}{10}$

444. $3\frac{4}{5}+6\frac{4}{5}$

445. $9\frac{2}{3}+1\frac{2}{3}$

446. $6\frac{9}{10}+8\frac{3}{10}$

447. $8\frac{4}{9}+2\frac{8}{9}$

Model Subtraction of Mixed Numbers

In the following exercises, use a model to find the difference. Draw a picture to illustrate your model.

448. $1\frac{1}{6}-\frac{5}{6}$

449. $1\frac{1}{8}-\frac{5}{8}$

Subtract Mixed Numbers with a Common Denominator

In the following exercises, find the difference.

450. $2\frac{7}{8}-1\frac{3}{8}$

451. $2\frac{7}{12}-1\frac{5}{12}$

452. $8\frac{17}{20}-4\frac{9}{20}$

453. $19\frac{13}{15}-13\frac{7}{15}$

454. $8\frac{3}{7}-4\frac{4}{7}$

455. $5\frac{2}{9}-3\frac{4}{9}$

456. $2\frac{5}{8}-1\frac{7}{8}$

457. $2\frac{5}{12}-1\frac{7}{12}$

Add and Subtract Mixed Numbers with Different Denominators

In the following exercises, write the sum or difference as a mixed number in simplified form.

458. $3\frac{1}{4}+6\frac{1}{3}$

459. $2\frac{1}{6}+5\frac{3}{4}$

460. $1\frac{5}{8}+4\frac{1}{2}$

461. $7\frac{2}{3}+8\frac{1}{2}$

462. $9\frac{7}{10}-2\frac{1}{3}$

463. $6\frac{4}{5}-1\frac{1}{4}$

464. $2\frac{2}{3}-3\frac{1}{2}$

465. $2\frac{7}{8}-4\frac{1}{3}$

Mixed Practice

In the following exercises, perform the indicated operation and write the result as a mixed number in simplified form.

466. $2\frac{5}{8}\cdot 1\frac{3}{4}$

467. $1\frac{2}{3}\cdot 4\frac{1}{6}$

468. $\frac{2}{7}+\frac{4}{7}$

469. $\frac{2}{9}+\frac{5}{9}$

470. $1\frac{5}{12}\div\frac{1}{12}$

471. $2\frac{3}{10}\div\frac{1}{10}$

472. $13\frac{5}{12}-9\frac{7}{12}$

473. $15\frac{5}{8}-6\frac{7}{8}$

474. $\frac{5}{9}-\frac{4}{9}$

475. $\frac{11}{15} - \frac{7}{15}$

476. $4 - \frac{3}{4}$

477. $6 - \frac{2}{5}$

478. $\frac{9}{20} \div \frac{3}{4}$

479. $\frac{7}{24} \div \frac{14}{3}$

480. $9\frac{6}{11} + 7\frac{10}{11}$

481. $8\frac{5}{13} + 4\frac{9}{13}$

482. $3\frac{2}{5} + 5\frac{3}{4}$

483. $2\frac{5}{6} + 4\frac{1}{5}$

484. $\frac{8}{15} \cdot \frac{10}{19}$

485. $\frac{5}{12} \cdot \frac{8}{9}$

486. $6\frac{7}{8} - 2\frac{1}{3}$

487. $6\frac{5}{9} - 4\frac{2}{5}$

488. $5\frac{2}{9} - 4\frac{4}{5}$

489. $4\frac{3}{8} - 3\frac{2}{3}$

Everyday Math

490. **Sewing** Renata is sewing matching shirts for her husband and son. According to the patterns she will use, she needs $2\frac{3}{8}$ yards of fabric for her husband's shirt and $1\frac{1}{8}$ yards of fabric for her son's shirt. How much fabric does she need to make both shirts?

491. **Sewing** Pauline has $3\frac{1}{4}$ yards of fabric to make a jacket. The jacket uses $2\frac{2}{3}$ yards. How much fabric will she have left after making the jacket?

492. **Printing** Nishant is printing invitations on his computer. The paper is $8\frac{1}{2}$ inches wide, and he sets the print area to have a $1\frac{1}{2}$-inch border on each side. How wide is the print area on the sheet of paper?

493. **Framing a picture** Tessa bought a picture frame for her son's graduation picture. The picture is 8 inches wide. The picture frame is $2\frac{5}{8}$ inches wide on each side. How wide will the framed picture be?

Writing Exercises

494. Draw a diagram and use it to explain how to add $1\frac{5}{8} + 2\frac{7}{8}$.

495. Edgar will have to pay $\$3.75$ in tolls to drive to the city.
ⓐ Explain how he can make change from a $\$10$ bill before he leaves so that he has the exact amount he needs.
ⓑ How is Edgar's situation similar to how you subtract $10 - 3\frac{3}{4}$?

496. Add $4\frac{5}{12} + 3\frac{7}{8}$ twice, first by leaving them as mixed numbers and then by rewriting as improper fractions. Which method do you prefer, and why?

497. Subtract $3\frac{7}{8} - 4\frac{5}{12}$ twice, first by leaving them as mixed numbers and then by rewriting as improper fractions. Which method do you prefer, and why?

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
model addition of mixed numbers common with a denominator.			
add mixed numbers with a common denominator.			
model subtraction of mixed numbers.			
subtract mixed numbers with a common denominator.			
add and subtract mixed numbers with different denominators.			